Dynamic Fusion Label Assignment Network for Remote Sensing Object Detection

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***Abstract*—Object detection in remote sensing images faces significant challenges posed by tiny objects, which are often overwhelmed by background noise. These tiny objects exhibit significant scale differences compared to larger objects, making it difficult to simultaneously consider both large and small objects during label assignment. In this article, a novel dynamic fusion label assignment network (DFLAN) is proposed to address these problems. First, to effectively extract features of tiny objects in background noise, we introduce a novel feature selection interactive pyramid network. Second, a novel dynamic fusion label assignment (DFLA) algorithm is developed, which achieves a collaborative approach to label assignment for both tiny and large objects. Finally, a new decoupling detection head is proposed to prevent task coupling from interfering with the already weak features of tiny objects. The proposed DFLAN method achieves state-of-the-art performance on two widely used datasets: DOTA-v1.0 (79.42% mAP), HRSC2016 (98.86% mAP), and DIOR-R (67.68% mAP).**

***Index Terms*—Feature fusion, label assignment, oriented object detection, task decoupling, Vision Mamba.**

I. Introduction

REMOTE sensing object detection involves identifying object entities while precisely determining their categorical classification and spatial positioning. Contrary to conventional visual scenes, remote sensing imagery is characterized by dense distributions of diminutive entities exhibiting pronounced heterogeneity across geometric dimensions, rotational angles, and topological arrangements. These objects are typically densely packed and may have low contrast against complex backgrounds, making their detection particularly challenging. When using neural networks to extract features from remote sensing objects, shallow networks often exhibit weak feature representation capabilities, leading to poor discrimination of small or occluded objects. Deeper networks, while enhancing feature representation, tend to suffer from reduced spatial localization accuracy due to the loss of fine-grained details during downsampling operations. Additionally, deeper networks introduce a higher computational burden, which may not be suitable for real-time remote sensing applications.

Horizontal object detectors have been widely adopted in general vision. Faster R-CNN [1] employs a two-stage design with a region proposal network (RPN) to achieve high accuracy but at a higher computational cost. YOLO [2] reformulates detection as a single regression for real-time inference, and DETR [3] introduces a Transformer-based end-to-end set-prediction framework that removes hand-crafted components and leverages global context. However, these horizontal detectors are primarily designed for natural images and axis-aligned objects, making them less effective in remote sensing scenarios where targets are often arbitrarily oriented and densely distributed.

To address the unique challenges of remote sensing, mainstream rotated object detectors such as RoI Trans [4], Oriented R-CNN [5], Oriented RepPoints [6], and R3Det [7] have incorporated the feature pyramid network (FPN) [8] to aggregate multiscale contextual information. However, the conventional fusion strategy in the FPN is limited: it primarily relies on a top-down pathway where enhanced semantic features are upscaled and then merged with shallow-layer counterparts. Although this methodology improves object detection accuracy, it presents inherent limitations when applied to remote sensing imagery. A primary limitation concerns FPN's additive fusion mechanism, which inadequately exploits cross-scale feature complementarity and amplifies vulnerability to background noise contamination. This issue becomes especially critical in RS imagery, as complex background textures are combined with spatially crowded object arrangements. In addition, the unidirectional cascade structure in the FPN constrains reciprocal information exchange, impairing multiscale contextual integration. This impedes accurate localization of spatially overlapping entities with fine-grained details, which demand high-resolution discrimination.

Recent advancements in addressing these constraints involve enhanced feature integration methodologies, including bidirectional FPNs (BiFPNs) [9] and attention-based feature aggregation methods [10]. Another line of work explores ensemble models, such as the multiple prediction mechanisms ensemble (MPME) [11], which improves detection in complex scenes through an ensemble of diverse feature

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pyramids and detection heads, leveraging backbone sharing for efficiency. These approaches aim to enhance cross-scale feature interactions, improve spatial localization precision, and reduce the impact of background noise. One promising direction for enhancing feature interactions is the introduction of visual Mamba [12], a technique that emphasizes dynamic, context-aware feature fusion. Visual Mamba, inspired by the flexible movement of a Mamba snake, enables more adaptive feature interactions across scales and spatial resolutions. This approach seeks to decouple spatial features from the global context and selectively integrate features from different levels depending on their task-relevant significance for object localization. By improving feature interactivity across scales and integrating context more effectively, Visual Mamba enhances the detector's ability to focus on fine-grained details. This achieves improved localization reliability for spatially compact, orientation-diverse, and closely clustered entities within RS datasets.

In addition to the problem of multiscale feature interaction, the significant variations in size and orientation of remote sensing objects pose substantial challenges to label assignment. Remote sensing images contain objects that are arbitrarily oriented and vary dramatically in size. This makes traditional label assignment strategies less effective, especially for small and rotated objects.

Mainstream rotated object detectors typically adopt the intersection over union (IoU) threshold strategy for label assignment. This process assigns positive and negative labels based on the IoU between the predicted and ground-truth (GT) bounding boxes. However, it has several limitations. First, IoU-based assignment tends to favor large objects, as their bounding boxes have a higher probability of overlapping with proposal boxes, leading to an imbalance where large objects dominate the training process. Second, IoU-based assignment struggles with highly rotated objects, as even a small angular mismatch can lead to a marked decrease in IoU values, causing unstable assignments and ineffective learning for such objects. Consequently, this imposes constraints on model efficacy, particularly for small and arbitrarily rotated objects.

To address this issue, Gaussian receptive field-based label assignment (RFLA) [13] employs Kullback–Leibler divergence (KLD) to directly measure the similarity between proposals and GTs. Instead of relying solely on IoU, this method models objects as Gaussian distributions, allowing for more flexible and robust label assignment. RFLA has demonstrated notable performance improvements in small-object detection, as it provides a smoother gradient for learning and reduces the strict dependence on high IoU overlap. However, in remote sensing scenarios, where objects exhibit significant scale variations and arbitrary orientations, RFLA faces limitations. Specifically, the Gaussian modeling may not accurately capture the extreme aspect ratios of some remote sensing objects, leading to suboptimal assignment results for large objects. Moreover, when objects are densely packed, the Gaussian distributions of multiple objects can overlap, causing ambiguity in label assignment. Addressing these challenges requires adaptive label assignment strategies that can dynamically adjust to the scale, orientation, and spatial distribution of remote sensing objects.

Beyond label assignment, another critical challenge in remote sensing object detection lies in the design of detection heads. Fully connected (FC) heads are generally better suited for classification tasks, as they can capture global feature representations and better distinguish between complete and partial objects. In contrast, convolutional (Conv) heads share convolutional kernels across all spatial locations, making them more appropriate for bounding box regression tasks [14]. In remote sensing images, where most objects are small, the effective features for classification and regression are often weak. Using a shared convolutional detection head can lead to feature entanglement, where the features useful for classification interfere with those required for regression. For small objects, this effect is particularly severe, as their already limited feature representation is further diluted by irrelevant information, effectively introducing strong noise into the learning process.

To mitigate this issue, recent works have explored decoupling classification and regression in the detection head. One promising direction is the task-aware head (TAH), which uses separate branches for classification and regression, each optimized for its respective task. Another approach involves adaptive feature selection mechanisms, such as dynamic convolutional filters or attention-based routing, which selectively refine the features passed to classification and regression branches. Additionally, methods like double-tower architectures introduce parallel pathways, ensuring that the information required for classification remains distinct from that needed for precise localization.

Despite these advancements, how to effectively decouple classification and regression while preserving computational efficiency remains an unresolved challenge in remote sensing object detection. Future work should focus on developing lightweight yet effective decoupled detection heads, potentially leveraging self-attention mechanisms or multi-scale feature refinement to enhance small-object detection performance without introducing excessive computational overhead.

To address the above issues, this article proposes a dynamic fusion label assignment network (DFLAN). The main contributions are as follows.

1) *Feature Selection Interaction Pyramid Network (FSIPN):* The FSIPN is composed of two key components: the feature selection module (FSM) and the feature interaction module (FIM). The FSM selectively enhances the features of objects based on Vision Mamba while suppressing irrelevant background interference. The FIM introduces a bottom-up feature interaction pathway, significantly enhancing the network's feature representation capabilities.
2) *Dynamic Fusion Label Assignment (DFLA):* DFLA is a novel label assignment strategy that combines IoU with KLD. It synergistically combines the advantages of KLD for labeling small objects and IoU for labeling large objects, demonstrating very good allocation performance for objects of various scales. Additionally, DFLA

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increases the number of positive samples, accelerating the model's convergence.

3) *Decouple Head (DHEAD):* The DHEAD integrates the large selective kernel attention and the depthwise separable convolutional sequentially to effectively decouple the regression from the classification. During training, the IoU threshold is dynamically increased to further enhance the performance of the DHEAD.

II. Related Work

A. Remote Sensing Object Detection Framework

Remote sensing object detection has advanced significantly in recent years. High-performance detectors typically follow the RCNN framework, combining an RPN with the RoI-wise CNN classification heads. This two-stage architecture has inspired adaptations to address challenges like arbitrary orientations, small and crowded objects, and complex backgrounds.

Traditional RCNN-based frameworks generate candidate regions in the first stage, followed by class prediction and coordinate refinement. The RoI Transformer [4] rotates horizontal anchor boxes during the first stage and extracts features for further regression and classification. The SCRDet [15] introduces an attention mechanism to suppress background noise, while the Oriented RCNN [5] and the Gliding Vertex [16] address instability caused by rotation angles with new encoding systems for oriented bounding boxes. Recent advancements, such as the LSKNet [17], ARC [18], and PKINet [19], improve backbone networks for better feature extraction in remote sensing tasks.

A notable advancement is QPDet [20], which uses radial–angular parameterization to improve boundary modeling and enhance detection accuracy without additional postprocessing. Similarly, the DRDet [21] introduces dual-angle rotated lines (DRLs), which adaptively fit the object's shape, reducing aliasing and improving orientation estimation, especially in dense aerial scenes.

Single-stage detectors like the RSDet [22], AOPG [23], and R3Det [7] simplify the process by deriving oriented bounding boxes directly from grid anchors. Transformer-based models, such as AO2-DETR [24], extend the use of Transformers in remote sensing tasks.

B. Multiscale Learning

A scale-invariant, log-stratified feature hierarchy enables robust localization across diverse scales. The PANet [25] strengthens the bottom-up path and refines top-down/bottom-up fusion for better multiscale features, while the BiFPN [26] assigns learnable per-level weights for balanced, adaptive fusion, improving robustness in complex scenes.

The RecursiveFPN [27] extends the FPN with recursive connections that iteratively refine feature maps at each scale, yielding better aggregation and representation across scales and benefiting small or densely packed objects. APDL [28] integrates attention-based pyramid dilated convolutions with residual and dense aggregations to focus on informative regions and effectively capture multilevel contextual information, improving robustness to the noisy backgrounds common in remote sensing images.

CFP [29] combines global feature regulation with local corner region capture to strengthen small-object detection, particularly in cluttered or occluded environments typical of remote sensing imagery. Addressing wide-scale variations, the multiple region proposal experts network (MRPENet) [30] employs parallel region proposal experts (RPEs) with varying receptive fields to specialize in detecting objects across different scale ranges, aided by an adaptive feature compensation module.

C. Label Assignment

Positive and negative sample distribution is crucial for training object detectors [31]. In recent years, numerous methods have been proposed to improve label assignment in object detection tasks. For instance, FreeAnchor [32] introduces a customized likelihood function to determine positive anchors in a detection-specific manner. PAA [33] uses a Gaussian mixture model (GMM) to model anchor distributions and classify positive/negative samples based on GMM centers. OTA [34] interprets the label assignment process through the lens of an optimal transport problem, aiming to identify the most efficient assignment strategy.

In addition, ATSS adapts the selection of pos/neg samples according to their statistical characteristics, such as their mean and variance, making the assignment process more flexible and robust. Meanwhile, AutoAssign [35] and IQDet [36] reweight and sample regions based on predicted IoU and confidence scores, thereby prioritizing high-quality regions. RFLA [13] uses Gaussian receptive fields for small-object detection and label assignment. Oriented RepPoints [6] improves performance by assessing the quality of points, enabling more effective label assignment. DCFL [37] achieves label assignment from coarse to fine by constructing coarse positive samples and representing objects with a dynamic GMM. These advancements collectively enhance label assignment, resulting in more accurate and efficient object detection models.

D. Decoupled Heads

In remote sensing object detection, particularly in challenging environments where small and rotated objects are prevalent, the decoupling of classification and localization tasks has become an important design strategy. Traditional detection frameworks typically share a single head for both classification and bounding box regression. However, this shared head can introduce feature confusion, especially when objects vary significantly in size or orientation. The two tasks—classification and regression—often require different feature representations, and sharing the same head may lead to suboptimal feature learning.

Recent works have introduced decoupled detection heads to mitigate feature interference between classification and regression. RepPoints [38] employs a decoupled regression mechanism for flexible bounding box prediction, which is effective for rotated objects. ATSS [31] and GFL [39] further separate the two tasks, yielding more stable and accurate predictions, especially for small or densely packed targets. Future research may explore adaptive decoupling strategies

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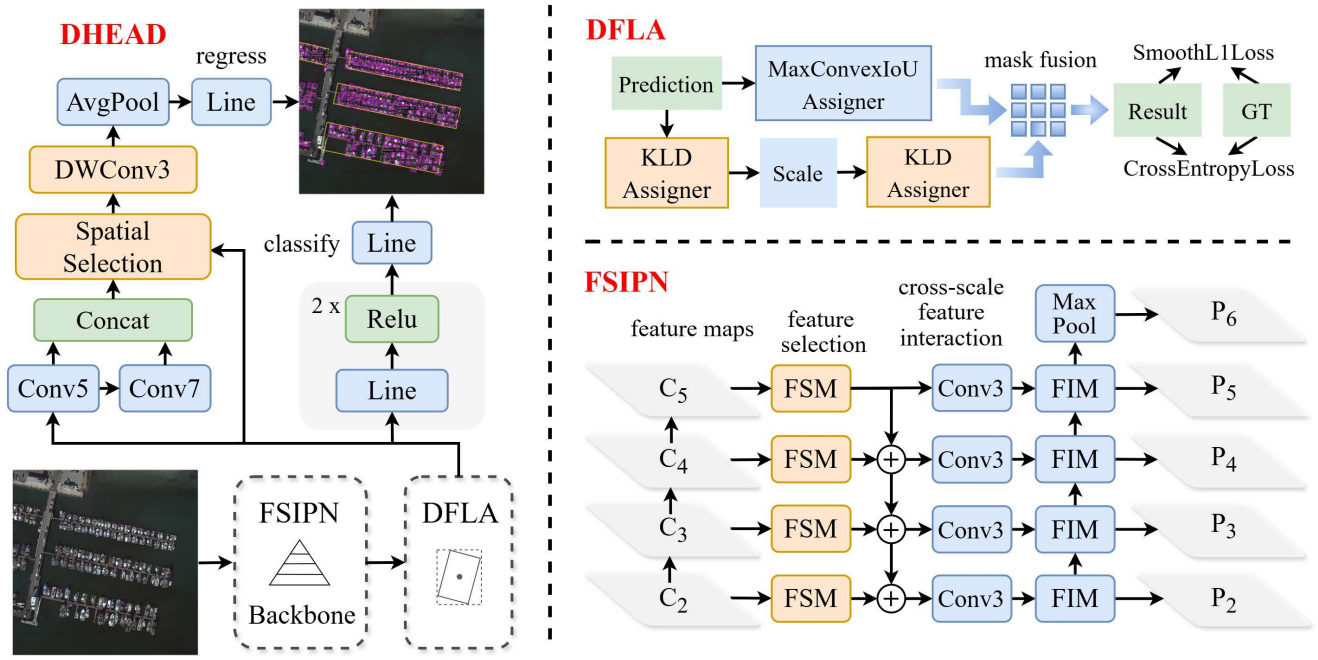


Fig. 1. Overall framework of the DFLAN. The FSIPN based on Vision Mamba is mainly used for remote sensing target information selection and cross-scale information communication. The DFLA is an algorithm designed based on KLD specifically for training-oriented region proposal networks (Oriented RPNs). The DHEAD is a lightweight detection head that separates regression and classification tasks.

that adjust according to object scale and orientation to further enhance remote sensing detection.

III. Method

A. Overall Architecture

The overall architecture of the DFLAN is shown in Fig. 1. Based on the Oriented R-CNN [5], the FSIPN, DFLA, and DHEAD are designed to optimize the network, while the backbone network is replaced with the LSKNet [17]. The FSIPN is designed to improve feature representation by selecting and interacting with multiscale features, enabling the network to better capture the characteristics of remote sensing objects. DFLA introduces a dynamic label assignment mechanism, incorporating components such as the MaxConvexIoU Assigner and KLD Assigner, to refine the training process of the Oriented RPN. The DHEAD decouples the detection task into regression and classification branches, allowing for more precise localization and classification of objects. Sections III-B–III-D will provide detailed descriptions of these three components.

B. Feature Selection Interaction Pyramid Networks

As shown in Fig. 1, the FSIPN is designed to enhance the multiscale feature fusion capability of object detectors while reducing complex background noise of remote sensing images. The FSIPN mainly consists of two modules: the FSM and the FIM.

As shown in Fig. 2, the architecture of the Mamba block consists of linear projection, depthwise convolution (DWConv), the SiLU activation function, and a structured state-space (SSM) model. The main path processes the input

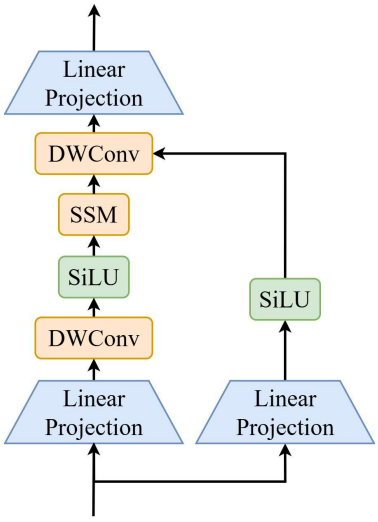


Fig. 2. Structure diagram of the Mamba block.

sequentially through these operations, while a parallel skip connection with linear projection and SiLU merges into the main path, enhancing feature extraction and information flow. Compared with conventional convolution or Transformer-based fusion, the Mamba block is suitable for remote sensing small-object detection. Its structured SSM enables long-range dependency modeling while maintaining linear computational complexity, which helps capture contextual cues for small objects surrounded by cluttered backgrounds. Moreover, the combination of DWConv and lightweight gating preserves fine-grained spatial details while filtering irrelevant noise, thereby providing more discriminative and noise-robust feature

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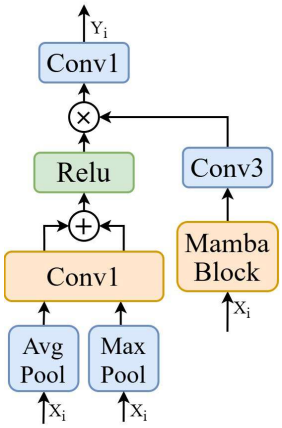


Fig. 3. Structure diagram of the FSM.

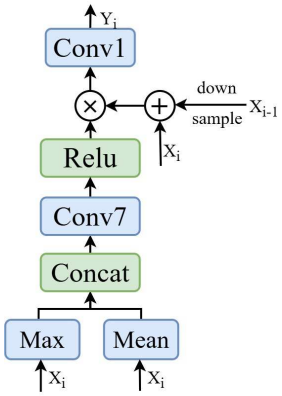


Fig. 4. Structure diagram of the FIM.

representations for small-scale objects. The detailed structures of the state-space FSM and FIM are shown in Figs. 3 and 4, respectively.

The backbone network consists of four stages, where C_i represents the output of the $i-1$ stage of the backbone network. In remote sensing images, most small objects are first perceived in the C3 layer, while some very small objects can be detected in the C2 layer. The features output by these low-level networks contain a lot of noise, so first, we use the FSM to selectively enhance object features while suppressing irrelevant background interference.

Although top-down information fusion can effectively aggregate high-level semantic features, it ignores the detailed information in low-level features, which can lead to inaccurate object positioning in complex scenarios. Therefore, we supplement the bottom-up information exchange path through FIM to improve the expression of detailed information.

The FSM module can be expressed by the following formula:

$$U_i = F_{\text{share}}^{1\times1}\left(\text{AvgPool}\left(X_i\right) + \text{MaxPool}\left(X_i\right)\right) \tag{1}$$

$$Y_i = F^{1\times1}\left(\text{ReLU}\left(U_i\right) \cdot F^{3\times3}\left(\text{MambaBlock}\left(X_i\right)\right)\right) \tag{2}$$

where the input feature map X_i first passes through average pooling (AvgPool) and maximum pooling (MaxPool) to extract global and local information, respectively. Then, the pooled features pass through shared convolution operations $F_{\text{share}}^{1\times1}$ to get the feature U_i, and the channel weights are generated through the ReLU activation function. In the second step, the features are first passed through the convolution $F^{3\times3}$ and then through the MambaBlock, as indicated in the formula. Finally, the weights are fused with the feature map via a dot product operation and a convolution operation $F^{1\times1}$ to enhance the important areas.

The FIM module can be expressed by the following formula:

$$U_i = F_{\text{share}}^{1\times1}\left(\text{AvgPool}(X_i) + \text{MaxPool}(X_i)\right) \tag{3}$$

$$Y_i = F^{1\times1}\left(U_i \cdot \text{DownSample}\left(X_{i-1}\right) + X_i\right) \tag{4}$$

Initially, the downsampling (DownSample) operation is applied to the low-level features X_{i-1} and then added to the feature map of this level. Subsequently, the maximum (Max) and average (Mean) operations are applied to the input X_i of the current layer, and the results are concatenated along the channel dimension to form the new feature U_i. Finally, this new feature representation is processed through a 7×7 convolution layer $F^{7\times7}$ followed by a ReLU activation function to generate the spatial weights. These weights are then fused with the feature map through a dot product operation and a 1×1 convolution layer $F^{1\times1}$ to enhance the flow of information across scales.

C. Dynamic Fusion Label Assignment

In the early stages of training, the model's predictions were not good enough. During object detector training, KLD benefits small objects [13], while IoU is better for large objects. Using either metric alone biases model optimization, hindering overall convergence. To address this, we propose a DFLA algorithm that integrates KLD and IoU synergistically. As shown in Algorithm 1, the DFLA consists of two stages: the first stage calculates a cost matrix using IoU to match the top-1 prediction box for each GT, and the second stage uses KLD to match the top K_1 and K_2 prediction boxes for each GT. The label assignment results from both stages are dynamically fused to produce the final assignment.

The input of DFLA includes the predicted horizontal bounding box set P and the GT bounding box set G. K_1 and K_2 are the number of top predicted boxes assigned to each GT when KLD is used as the metric for assignment. P_{thr} and N_{thr} are the positive sample IoU threshold and negative sample IoU threshold, respectively. L is the scale reduction ratio of predicted bounding boxes.

In the first stage, DFLA uses the MaxConvexIoU Assigner to obtain the auxiliary assignment result R_a. The positive samples assigned in R_a are of very high quality, but the number is relatively small. Specifically, the horizontal predicted bounding boxes P are converted into point set representations of the directed bounding boxes $\bar{P}$. The angle representations of the rotated bounding boxes G are then converted into quadrilateral representations $\bar{G}$. Then, DFLA calculates the convex hull IoU between the candidate boxes $\bar{P}$ and GT boxes $\bar{G}$. Based on

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the ConvexIoU values, it marks predicted boxes as positive samples if the IoU exceeds P_{thr} or as negative samples if the IoU falls below N_{thr}.

Algorithm 1 Dynamic Fusion Label Assignment

Input: P: a set of predicted horizontal bounding boxes
G: a set of ground truth(GT) oriented bounding boxes
P_{thr}: positive sample IoU threshold
N_{thr}: negative sample IoU threshold
L: scale reduction ratio of predicted bounding boxes
K_1, K_2: the number of boxes assigned to each GT

$\bar{P} = hbox2pbox(P)$
$\bar{G} = rbox2qbox(G)$
for each predicted box $p_i \in \bar{P}$ **do**
  for each ground truth box $g_j \in \bar{G}$ **do**
    ConvexIoU: iou_{ij} = ConvexIoU(p_i, g_j)
    if $iou_{ij} >= P_{thr}$ **then**
      Mark p_i as a positive sample in R_a
    end if
    if $iou_{ij} < N_{thr}$ **then**
      Mark p_i as a negative sample in R_a
    end if
  end for
end for
KLD-based IoU matrix 1: $m_{ij} = KLD(P_j, G_i)$
for $iou_i \in m_{ij}$ **do**
  Mark $\{P_j \mid iou_{ij} \in TopK_1(iou_i)\}$ positive in R:
  Mark $\{P_j \mid iou_{ij} \notin TopK_1(iou_i)\}$ negative in R:
end for
L times scale reduction: $\tilde{P} \leftarrow P$
KLD-based IoU matrix 2: $\tilde{m}_{ij} = KLD(\tilde{P}_j, G_i)$
for $iou_i \in \tilde{m}_{ij}$ **do**
  Mark $\{P_j \mid iou_{ij} \in TopK_2(iou_i)\}$ positive in $\tilde{R}$:
  Mark $\{P_j \mid iou_{ij} \notin TopK_2(iou_i)\}$ negative in $\tilde{R}$:
end for
$mask_1 = (R > 0)$
$mask_2 = (R <= 0)$
$R_b = mask_1 * R + mask_2 * \tilde{R}$
$mask = (R_a > 0)\&(R_b <= 0)$
$A =\sim \text{mask} * R_a + \text{mask} * R_b$
return A

In the second stage, DFLA calculates the cost matrix m_{ij}, which represents the KLD-based IoU between the predicted bounding box P_j and the GT bounding box G_i. For each GT bounding box G_i, the top K_1 predictions with the largest KLD-based IoU values in m_{ij} are marked as positive samples in R, while the rest are marked as negative samples. Next, after scaling down the predicted bounding boxes L times, a new cost matrix $\tilde{m}_{ij}$ is recalculated, and the top K_2 predictions are marked as positive samples in $\tilde{R}$. Finally, a mask fusion operation is performed to obtain R_b. Specifically, positive samples in R are selected, and for negative samples in R, the assignment results from $\tilde{R}$ are chosen.

Ultimately, R_a is used as a high-quality auxiliary assignment result to supplement the assignment result R_b and generate the final assignment result R. The DFLA uses KLD and IoU as metrics to dynamically combine the results of two label assignment stages. KLD is more favorable for label assignment to dense small objects, while IoU supplements high-quality positive samples, improving both the optimization speed and effectiveness.

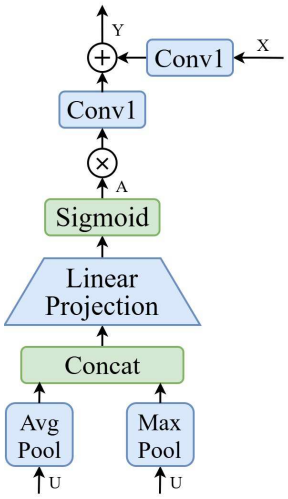


Fig. 5. Structure diagram of spatial selection.

D. Decouple Head

Sharing the convolutional head for classification and regression introduces task coupling, which further disrupts the already weak features of small objects. The FC layer performs exceptionally well in classification problems, whereas the convolutional layer is more effective for regression tasks. Therefore, we designed the decouple head (DHEAD) to leverage the strengths of both.

As shown in Fig. 1, we proposed a lightweight convolution task branch, whose structure draws on the large selection kernel attention in the LSKNet [17]. 5×5 DWConv $F^{5\times5}$ and 7×7 DWConv $F^{7\times7}$ are used to extract information from the input feature map X in different receptive fields

$$U = \left[F_{dw}^{5\times5}(X); F_{dw}^{7\times7}(X)\right]. \tag{5}$$

Fig. 5 shows the structure of spatial selection. The extracted features U are average-pooled (P_{avg}) and max-pooled (P_{max}), respectively, and then concatenated to generate the attention weight A

$$A = \sigma\left(F_{2\to N}^{1\times1}\left(\left[P_{\text{avg}}(U); P_{\text{max}}(U)\right]\right)\right) \tag{6}$$

where $F_{2\to N}^{1\times1}$ means using 1×1 convolution to reproject the channel 2 to N, and σ is the Sigmoid activation function. The final implementation of space selection is as follows:

$$Y = F_{N\to M}^{1\times1}(A \cdot X) + F_{N\to M}^{1\times1}(X) \tag{7}$$

where A represents the attention weight and X represents the input feature. The result is then passed through the mapping $F_{N\to M}^{1\times1}$ and added to the residual projection $F_{N\to M}^{1\times1}(X)$.

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Following closely is the depthwise separable convolution module (DSCM) we have designed, which is composed of two depthwise separable convolutions connected in series:

$$Y = F_{N\to M}^{1\times 1}\left(F_{dw}^{3\times 3}\left(F_{M\to N}^{1\times 1}(X)\right)\right) \tag{8}$$

where $F_{N\to M}^{1\times 1}$ means using 1×1 convolution to reproject the channel N to M, $F_{M\to N}^{1\times 1}$ means using 1×1 convolution to reproject the channel M to N, and $F_{dw}^{3\times 3}$ represents a DWConv of size 3×3. Finally, we finally obtain the regression coordinates of the bounding box through AvgPool and linear mapping (Line).

In the classification branch, we use the Line and ReLU activation functions to obtain the final classification result. This design effectively improves task performance while ensuring computing efficiency. We designed a dynamic adjustment strategy during training. As training progresses, the IoU threshold gradually increases, so that the model focuses more strictly on high-quality positive samples and more accurate bounding box regression. Threshold adjustment requires minimal computation, incurring almost no extra cost.

IV. Experiments

A. Datasets

The DOTA-v1.0 dataset comprises a total of 2806 high-resolution aerial imagery samples, annotated with 188 282 object instances across 15 frequently encountered object classes. These include aircraft-related targets (plane/PL and helicopter/HC), transportation infrastructure (bridge/BR, roundabout/RA, and harbor/HA), sports facilities [baseball diamond/BD, tennis court/TC, basketball court/BC, and soccer-ball field (SBF)], vehicles (small vehicle/SV, and large vehicle/LV), maritime vessels (ship/SH), industrial structures (storage tank/ST), and specialized terrain features ground track field (GTF) as well as recreational pools (swimming pool/SP).

HRSC2016 is a remote sensing dataset specifically designed for ship detection. We use 436 images from the training set and 181 images from the validation set for training and evaluate on 444 images from the testing set.

DIOR-R comprises 23 463 carefully curated remote sensing images, each with a fixed dimension of 800×800 pixels and spatial resolution varying from 0.5 to 30 m. The dataset is partitioned into a training and validation subset containing 11 725 images and a testing subset with 11 738 images, enabling comprehensive evaluation of object detection algorithms.

B. Implementation Details

To ensure fairness and reliability, the dataset configurations align with those used by mainstream methods [5], [17], [19]. For the Oriented RPN, the number of proposals is set to 2000. For the DHEAD, nonmaximum suppression is applied to the predicted oriented bounding boxes of each object category with a category probability greater than 0.05, using an IoU threshold of 0.1. On the DOTA-v1.0 and DIOR-R datasets, the AdamW optimizer is used to train the DFLAN for 12 epochs, starting with an initial learning rate of 1e-4. The learning rate is reduced by a factor of ten after the 8 and 11 epochs. For the HRSC2016 dataset, the DFLAN is trained for 36 epochs using AdamW, with an initial learning rate of 2e-4, and the learning rate is decreased by a factor of ten after the 24th and 33rd epochs.

C. Comparison With State-of-the-Arts

1) Performance Comparison on the DOTA-v1.0 Dataset: As shown in Table I, the DFLAN is compared with 20 classic methods on the DOTA-v1.0 dataset, surpassing all existing methods with an mAP of 79.42%. After introducing random rotation data augmentation, the mAP is further improved to 79.68%. Specifically, the DFLAN performs well in several difficult categories: the detection accuracy of bridge (BD) reaches 85.26%; the accuracy of GTF is 79.78%; and the SBF achieves a rate of 69.79%.

As shown in Fig. 6, our method demonstrates superior performance when detecting large objects in field-related categories, such as ships, bridges, buildings, and harbors. This highlights DFLAN's robustness in scenarios with vast spatial scales and intricate object details. At the same time, it maintains high detection accuracy for smaller objects, such as small vehicles and large vehicles, which are typically more challenging due to their reduced spatial footprint and higher susceptibility to background noise.

As shown in Fig. 7, the visualization results on the DOTA-v1.0 dataset further compare the DFLAN with the LSKNet. It can be observed that the LSKNet suffers from missed detections, inaccurate localization, and category confusion in dense or complex scenes, such as vehicles, aircraft, and sports fields. In contrast, the DFLAN significantly alleviates these issues by producing more accurate bounding boxes and reducing false alarms. These improvements demonstrate the effectiveness of our model in handling crowded layouts and complex backgrounds in large-scale remote sensing imagery.

2) Performance Comparison on the HRSC2016 Dataset: As shown in Table II, the DFLAN achieved 90.71% and 98.86% mAP based on the PASCAL VOC 2007 and 2012 evaluation standards, respectively, both of which set new performance records for the dataset. Compared with 90.60% of the RTMDet under the VOC2007 standard and 98.46% of LSKNet-S under the VOC2012 standard, DFLAN's leading advantage has been further expanded. In addition, with 35.4M parameters, the DFLAN significantly outperforms high-parameter models such as the DRN (41.1M) and O-RCNN (41.1M), verifying the algorithm's ability to balance detection accuracy, computational efficiency, and model complexity.

As shown in Fig. 8, the visualization results on the HRSC2016 dataset are presented. The DFLAN demonstrates exceptional capability in multiscale ship detection, accurately identifying objects of varying sizes. The visualization results clearly illustrate the superior performance of our DFLAN on the ship detection task.

3) Performance Comparison on the DIOR-R Dataset: As shown in Table III, the DFLAN achieves the best overall performance on the DIOR-R dataset, reaching an mAP of 67.68%. After introducing random rotation data augmentation, the result is further improved to 68.23%. These results verify the effectiveness of the DFLAN in addressing the challenges of multiclass, multiscale, and arbitrarily oriented objects, and

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TABLE I

COMPARISONS WITH STATE-OF-THE-ART METHODS ON THE DOTA-V1.0 OBB TASK. ALL EXPERIMENTAL RESULTS ARE BASED ON SINGLE-SCALE TRAINING AND TESTING, AND † INDICATES THAT RANDOM ROTATION DATA AUGMENTATION IS ENABLED. BEST IS IN BOLD

Method	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP_{50} ↑
DETR-based:																
AO^2-DETR [24]	87.99	79.46	45.74	66.64	78.90	73.90	73.30	90.40	80.55	85.89	55.19	63.62	51.83	70.15	60.04	70.91
O^2-DETR [40]	86.01	75.92	46.02	66.65	79.70	79.93	**89.17**	90.44	81.19	76.00	56.91	62.45	64.22	65.80	58.96	72.15
ARS-DETR [41]	86.61	77.26	48.84	66.76	78.38	78.96	87.40	90.61	82.76	82.19	54.02	62.61	72.64	72.80	64.96	73.79
One-stage:																
DAL [42]	88.68	76.55	45.08	66.80	67.00	76.76	79.74	90.84	79.54	78.45	57.71	62.27	69.05	73.14	60.11	71.44
S^2A-Net [43]	89.11	82.84	48.37	71.11	78.11	78.39	87.25	90.83	84.90	85.64	60.36	62.60	65.26	69.13	57.94	74.12
TIR-Net [44]	89.26	82.42	53.57	72.23	79.15	78.87	86.96	90.87	84.25	84.76	65.43	68.50	67.60	70.35	54.22	75.23
KLD [45]	88.91	83.71	50.10	68.75	78.20	76.05	84.58	89.41	86.15	85.28	63.15	60.90	75.06	71.51	67.45	75.28
O-RepPoints [6]	87.02	83.17	54.13	71.16	80.18	78.40	87.28	**90.90**	85.97	86.25	59.90	70.49	73.53	72.27	58.97	75.97
R^3Det [7]	89.80	83.77	48.11	66.77	78.76	83.27	87.84	90.82	85.38	85.51	65.67	62.68	67.53	78.56	**72.62**	76.47
Two-stage:																
CenterMap [46]	89.02	80.56	49.41	61.98	77.99	74.19	83.74	89.44	78.01	83.52	47.64	65.93	63.68	67.07	61.59	71.59
SCRDet [15]	**89.98**	80.65	52.09	68.36	68.36	60.32	72.41	90.85	**87.94**	**86.86**	65.02	66.68	66.25	68.24	65.21	72.61
RoI Trans. [4]	88.65	82.60	52.53	70.87	77.93	76.67	86.87	90.71	83.83	82.51	53.95	67.61	74.67	68.75	61.03	74.61
G.V. [16]	89.64	85.00	52.26	77.34	73.01	73.14	86.82	90.74	79.02	86.81	59.55	**70.91**	72.94	70.86	57.32	75.02
O-RCNN [5]	89.46	82.00	54.79	76.12	78.25	83.00	70.72	89.86	80.42	84.68	63.97	67.69	74.94	68.84	52.28	75.87
ReDet [47]	88.79	82.64	53.97	74.00	78.13	84.06	88.04	90.89	87.78	85.75	61.76	60.39	75.96	68.07	63.59	76.25
DRDet [21]	**88.98**	82.63	49.63	64.14	**81.77**	83.99	88.25	**90.90**	86.86	85.89	58.80	66.65	74.88	78.14	64.58	76.40
RPG-Ori-R [48]	89.90	83.92	53.72	70.99	78.20	83.42	88.04	90.88	86.72	85.94	67.21	68.47	75.66	72.10	62.38	77.17
ARC [18]	89.40	82.48	55.33	73.88	79.37	84.05	88.06	**90.90**	86.44	84.83	63.63	70.32	74.29	71.91	65.43	77.35
LSKNet-S [17]	89.65	84.74	55.51	78.18	79.60	84.98	88.34	**90.90**	86.48	86.72	65.43	69.58	77.12	73.72	57.37	77.89
PKINet-S [19]	89.72	84.20	55.81	77.63	80.25	84.45	88.12	90.88	87.57	86.07	66.86	70.23	**77.47**	73.62	62.94	78.39
DFLAN(ours)	89.56	**85.26**	54.78	**79.78**	79.36	85.17	88.50	**90.90**	86.54	86.32	**69.79**	67.75	77.35	79.45	70.85	79.42
DFLAN†(ours)	89.63	85.17	**55.84**	78.92	79.84	**85.66**	88.43	90.88	**87.94**	86.84	69.19	68.02	77.40	**81.89**	69.51	**79.68**

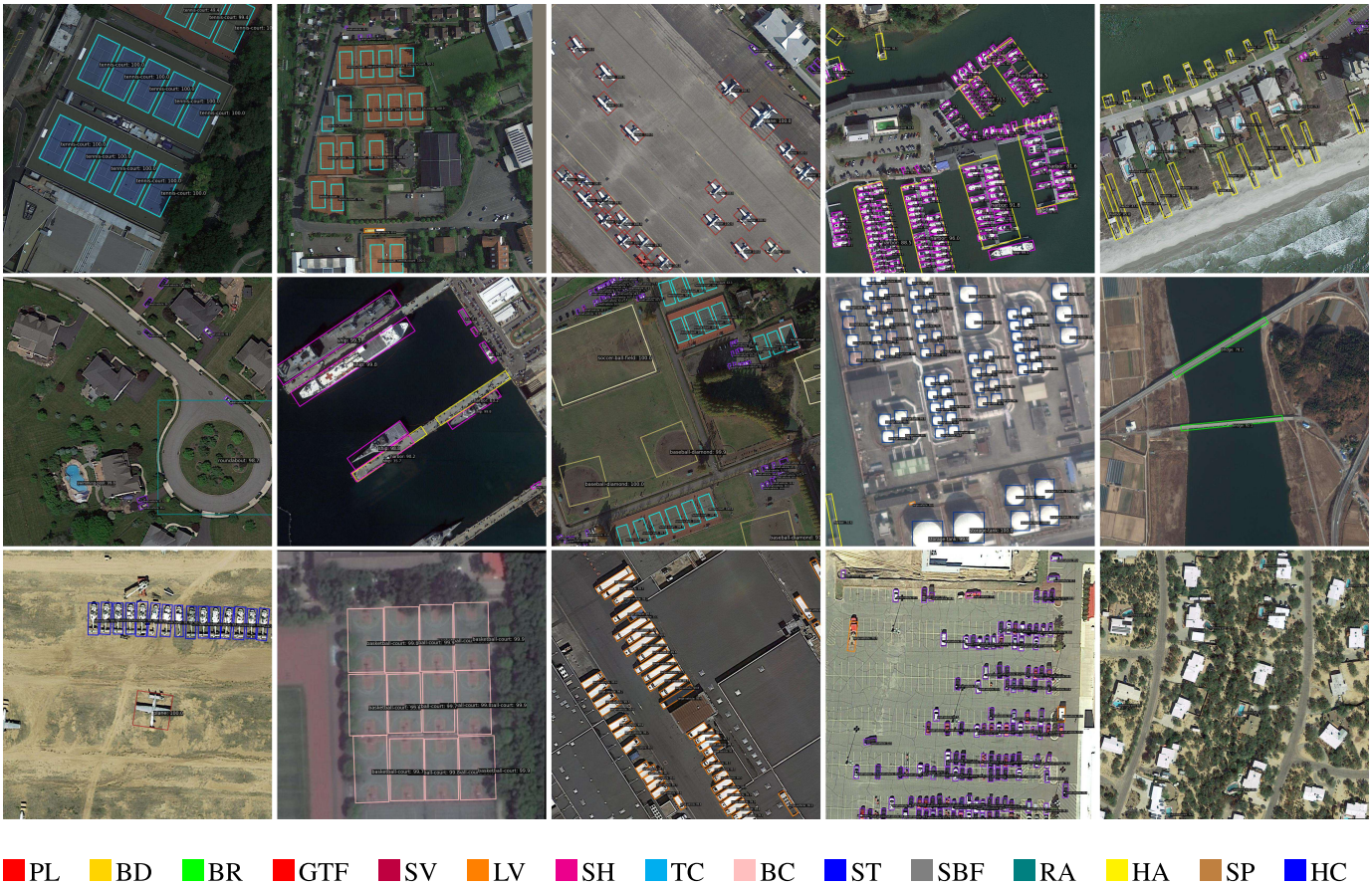


Fig. 6. Visualization of detection results on the DOTA-v1.0 dataset.

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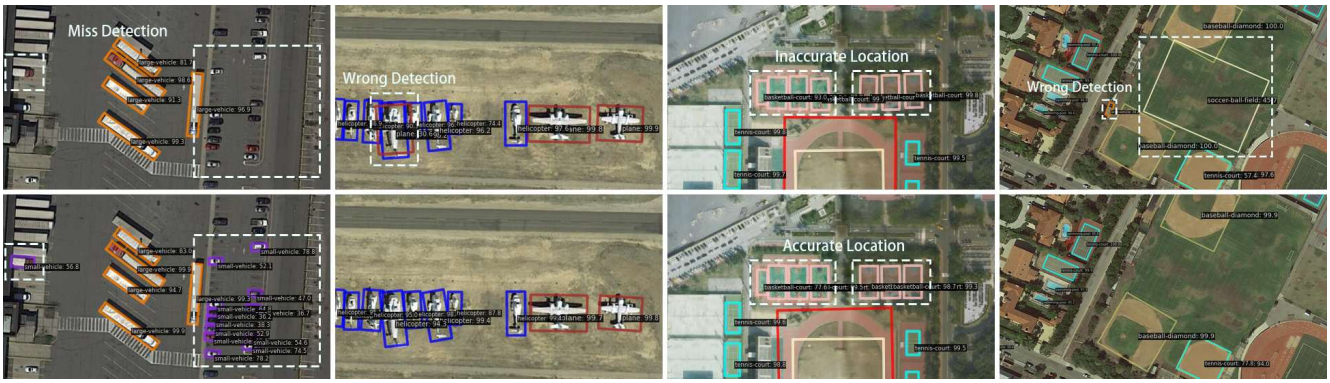


Fig. 7. Comparison of visualization results on the DOTA-v1.0 dataset. (Top) LSKNet. (Bottom) Our DFLAN.

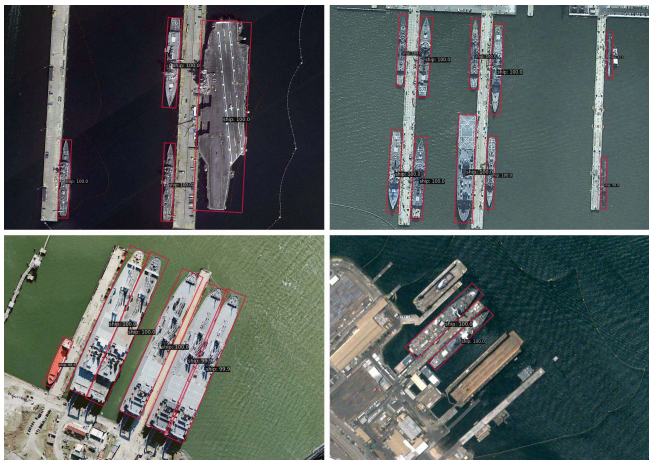


Fig. 8. Visualization of detection results on the HRSC dataset.

TABLE II
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE HRSC2016 DATASET

Method	$mAP_{50}(07)\uparrow$	$mAP_{50}(12)\uparrow$	$\#P\downarrow$
DRN [49]	-	92.70	-
CenterMap [46]	-	92.80	41.1M
Rol Trans. [4]	86.20	-	55.1M
G.V. [16]	88.20	-	41.1M
R3Det [7]	89.26	96.01	41.9M
GWD [50]	89.85	97.37	47.4M
S^2ANet [43]	90.17	95.01	38.6M
AOPG [23]	90.34	96.22	-
ReDet [47]	90.46	97.63	31.6M
O-RCNN [5]	90.50	97.60	41.1M
RTMDet [51]	90.60	97.10	52.3M
LSKNet-S [17]	90.65	98.46	**31.0M**
DFLAN(ours)	**90.71**	**98.86**	35.4M

demonstrate its strong generalization capability in complex remote sensing scenarios.

As shown in Fig. 9, the visualization results on the DIOR-R dataset compare our DFLAN with the LSKNet. It can be observed that the DFLAN produces more precise and complete detections in scenes with dense and arbitrarily oriented objects, such as airplanes, storage tanks, and harbors. In particular, our method reduces missed detections and overlapping bounding boxes, showing stronger robustness in complex backgrounds and crowded layouts.

TABLE III
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE DIOR-R DATASET. † INDICATES THAT RANDOM ROTATION DATA AUGMENTATION IS ENABLED

Method	$mAP_{50}\uparrow$	Method	$mAP_{50}\uparrow$
RetinaNet [52]	57.60	FR-O [1]	57.60
G.V. [16]	60.06	S^2ANet [43]	61.40
RoI Trans. [4]	63.87	AMMBA [53]	64.60
LSKNet-S	65.90	O-RepPoints [6]	66.71
DCFL [37]	66.80	PKINet-S [19]	67.03
DFLAN(ours)	67.68	DFLAN†(ours)	**68.23**



Fig. 9. Comparison of visualization results on the DIOR-R dataset. (Top) LSKNet. (Bottom) Our DFLAN.

D. Ablation Studies

We conducted a series of ablation experiments on the FSIPN, DFLA, and DHEAD. These experiments were conducted on the DOTA-v1.0 dataset with the LSKNet as the backbone. Since the DOTA test server does not return the mAP for small objects separately, we use the average mAP of the five categories (SV, LV, SH, ST, and HC) that cover most of the small objects as the mAP for small objects (mAP_s).

1) Ablation of Different Modules: As shown in Table IV, the LSKNet is used as a baseline without incorporating the modules in our DFLAN, achieving a baseline mAP of 77.89% and a small-object precision (mAP_s) of 79.40%. After adding the FSIPN module, the results increased to 78.30% mAP and 80.89% mAP_s, which highlights its effectiveness in enhancing

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TABLE IV

ON THE DOTA-V1.0 DATASET, ABLATION EXPERIMENTS ARE CONDUCTED FOR EACH PROPOSED MODULE. WE CHOOSE THE LSKNET AS THE BASELINE, AND ✓ INDICATES THE USE OF THIS MODULE

Method	FSIPN	DFLA	DHEAD	mAP_{50} ↑	mAP_s ↑
Baseline				77.89	79.40
DFLAN	✓			78.30	80.89
DFLAN		✓		78.68	80.82
DFLAN			✓	78.55	80.61
DFLAN	✓	✓		78.98	81.55
DFLAN	✓	✓	✓	**79.42**	**82.04**

TABLE V

ON THE DOTA-V1.0 DATASET, WE CONDUCTED ABLATION EXPERIMENTS ON VARIOUS COMPONENTS OF THE FSIPN. FSM* INDICATES THE USE OF FSM COMPONENTS OTHER THAN MAMBA BLOCK, AND ✓ INDICATES THE USE OF THIS MODULE

Mamba Block	FSM*	FIM	mAP_{50} ↑	mAP_s ↑
			77.89	79.40
✓			78.06	79.93
✓	✓		78.15	80.25
		✓	78.09	79.60
✓	✓	✓	**78.30**	**80.89**

feature selection and enabling cross-scale feature interaction, particularly for small objects. Introducing the DFLA module further improved the results to 78.98% mAP and 81.55% mAP_s, demonstrating its ability to refine bounding box assignment and provide more reliable supervision for tiny targets. Finally, with the addition of the DHEAD module on top of FSIPN and DFLA, the overall performance reached 79.42% mAP and 82.04% mAP_s, achieving gains of +1.53% mAP and +2.64% mAP_s over the baseline. These results distinctly highlight the unique contributions of each module, their combined synergistic effects, and provide robust validation for the overall effectiveness of our DFLAN.

2) Analyzing the Roles of Different Components of the FSIPN: As shown in Table V, we conducted ablation experiments to investigate the contributions of different components within the FSIPN. When only the baseline network is used, the model achieves 77.89% mAP and 79.40% mAP_s. Introducing the Mamba block alone raises the results to 78.06% mAP and 79.93% mAP_s, indicating that this module effectively suppresses background interference and enhances discriminative features of tiny targets. When combined with other FSM components and the FIM, the performance further increases to 78.30% mAP and 80.89% mAP_s, demonstrating a cumulative gain of +0.41% mAP and +1.49% mAP_s compared with the baseline.

It is worth noting that the improvement on mAP_s (small-object mean average precision, defined as the average precision of SV, LV, SH, ST, and HC categories) is more significant than that on mAP. This observation highlights that Vision Mamba is particularly effective for small-object detection in remote sensing imagery, where fine-grained details are easily overwhelmed by background noise.

3) Analysis of Parameter L of DFLA: As shown in Table VI, the dynamic label assignment algorithm is significantly sensitive to the prediction box scale reduction ratio

TABLE VI

EFFECT OF DIFFERENT L IN DYNAMIC FUSION LABEL ASSIGNMENT ALGORITHM ON THE DOTA-V1.0 DATASET

L	0.75	0.80	0.85	0.90	0.95	1.0
mAP_{50} ↑	77.88	78.13	78.46	**78.68**	78.37	77.95

TABLE VII

EXPERIMENTAL RESULTS ON THE DOTA-V1.0 DATASET USING DIFFERENT METRICS AT DIFFERENT STAGES OF LABEL ASSIGNMENT

1st stage	2st stage A	2st stage B	mAP_{50} ↑	mAP_s ↑
IoU	-	-	77.89	79.40
IoU	IoU	IoU	78.26	79.52
IoU	KLD	IoU	78.35	80.28
IoU	IoU	KLD	78.15	79.89
IoU	KLD	KLD	**78.68**	**80.82**
KLD	IoU	IoU	77.83	79.35
KLD	KLD	KLD	76.69	78.89

L: when $L = 0.90$, the model achieves the optimal mAP of 78.68%, an improvement of 0.73%. This phenomenon shows that moderate scale reduction (L‘!1) improves the model performance by enhancing the recall rate of small target detection, while excessive reduction (such as 77.88% when $L = 0.75$) leads to a decrease in accuracy due to the introduction of noise samples, and the overly conservative $L = 1.0$ performs poorly due to insufficient adaptation to the change in target scale. It is worth noting that when L is in the range of [0.85, 0.95], the mAP fluctuates less (78.46%–78.68%), reflecting the robustness of the algorithm within a reasonable parameter range.

4) Differentiated Metrics of DFLA: It is worth noting that the improvement on mAP_s (small-object mean average precision, defined as the average precision of SV, LV, SH, ST, and HC categories) is more significant than that on mAP. This observation highlights that Vision Mamba is particularly effective for small-object detection in remote sensing imagery, where fine-grained details are easily overwhelmed by background noise. As shown in Table VII, when IoU is uniformly used as the matching criterion in the first stage, the model achieves 77.89% mAP and 79.40% mAP_s. If the KLD–IoU hybrid strategy is introduced only in the second stage (such as IoU is used in A and KLD is used in B), the performance is improved to 78.35% mAP and 80.28% mAP_s. Further optimization found that by dynamically adjusting the metrics of each stage (such as IoU in the first stage + KLD in the second stage + IoU in B), the highest scores of 78.68% mAP and 80.82% mAP_s can be achieved, which are 0.79% and 1.42% higher than the baseline, respectively. It is worth noting that the uniform use of KLD in all stages (the last row) causes the performance to drop to 76.69% mAP and 78.89% mAP_s, which verifies the irreplaceable role of IoU in initial matching. These results indicate that while IoU ensures stable initial matching, the incorporation of KLD provides additional benefits for small objects, leading to consistent improvements in mAP_s across different settings.

5) DHEAD Component Configuration: By adjusting the number of depthwise separable convolution (DWConv3), the spatial selection module (Spatial Selection), and the linear

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Fig. 10. Proposals generated on the DOTA-v1.0 dataset after training the model using the DFLA algorithm.

TABLE VIII
COMPUTATIONAL COMPLEXITY AND EXPERIMENTAL RESULTS OF DIFFERENT NUMBERS OF COMPONENTS OF THE DECOUPLED HEAD ON THE DOTA-V1.0 DATASET

DWConv3	Spatial Selection	Line and Relu	mAP_{50} ↑	FLOPs↓
2	1	2 → 1	77.39	352G
2	1 → 0	2	78.18	318G
2 → 1	1	2	78.25	**302G**
2	1	2	**78.55**	354G
2 → 3	1	2	78.54	406G
2	1 → 2	2	77.64	417G
2	1	2 → 3	78.52	356G

layer + ReLU (Line and ReLU), an effective balance can be achieved between detection accuracy and computational complexity. As shown in Table VIII, when retaining two layers of DWConv3 and gradually reducing the number of spatial selection modules (1 → 0), the mAP slightly decreases from 78.55% to 78.18%, but FLOPs significantly decreases to 318G; conversely, adding spatial selection modules (1 → 2) causes FLOPs to surge to 417G, while the mAP decreased from 78.55% to 77.64%. In addition, the number of linear layers + ReLU has a significant impact on computational complexity: when it is reduced from two layers to one layer, FLOPs decrease by 14%, but the mAP loses 1.16%. Comprehensive analysis shows that the optimal configuration is to retain two layers of DWConv3, one layer of spatial selection module, and two layers of linear layers + ReLU (mAP = 78.55%, FLOPs = 354G), which ensures detection accuracy while achieving effective utilization of computing resources.

E. Analysis

1) Proposals Visualization: As shown in Fig. 10, the candidate proposals generated by our DFLAN model on the DOTA-v1.0 dataset demonstrate the effectiveness of the proposed DFLA algorithm in training RPNs. The green bounding boxes highlight the model's ability to capture objects across diverse scales and orientations, including small vehicles, densely packed residential structures, and large-scale scenes like parking lots and water bodies. Key observations include the following.

1) *Multiscale Adaptability:* Proposals accurately localize both tiny objects (e.g., cars in parking lots) and large objects (e.g., bridges), validating the FSIPN module's capability to enhance features at multiple scales while suppressing background noise.
2) *Orientation-Aware Detection:* Rotated objects such as ships and tennis courts are consistently detected with precise alignment, showcasing the synergy between KLD-based label assignment and IoU-guided refinement in DFLA.
3) *Robustness to Complex Scenes:* In cluttered environments (e.g., dense residential areas), the model generates clean proposals without false positives, highlighting the FSM module's feature selection mechanism. These results confirm that DFLA dynamically balances label assignment for small and large objects, enabling efficient training and improving subsequent detection accuracy.

2) Feature Map Visualization: As shown in Fig. 11, the feature map visualizations of the FPN (top) and the FSIPN (bottom) on the DOTA-v1.0 dataset are illustrated. The top image shows the feature map output by FPN, which provides basic multiscale feature fusion but retains noticeable background noise. In contrast, the bottom image highlights the enhancement effect of the FSIPN, where the feature selection and interaction modules significantly strengthen target-related features while suppressing irrelevant background information. Notably, the FSIPN enhances regions corresponding to the vertices of rotated polygons, aiding bounding box regression, and suppresses noise from trees, roads, or empty areas, thereby emphasizing object features such as ships and vehicles for more accurate classification and regression.

3) Training Speed Improvement: On the HRSC2016 dataset, we use different training schedules and train our model. Specifically, the 1× schedule corresponds to 12 epochs,

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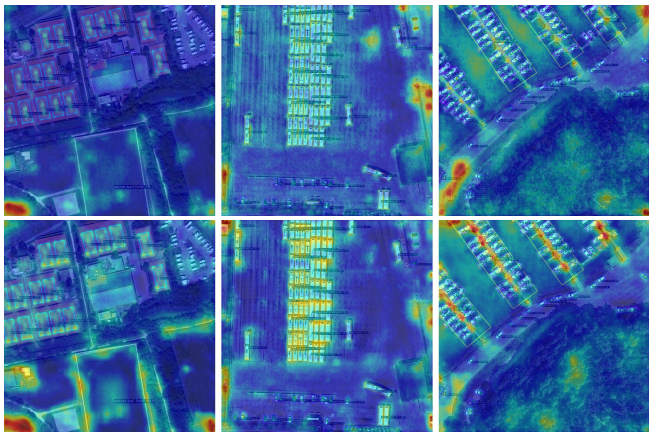


Fig. 11. Visualization of feature maps from the FSIPN and the FPN on the DOTA-v1.0 dataset. (Top) FPN. (Bottom) FSIPN.

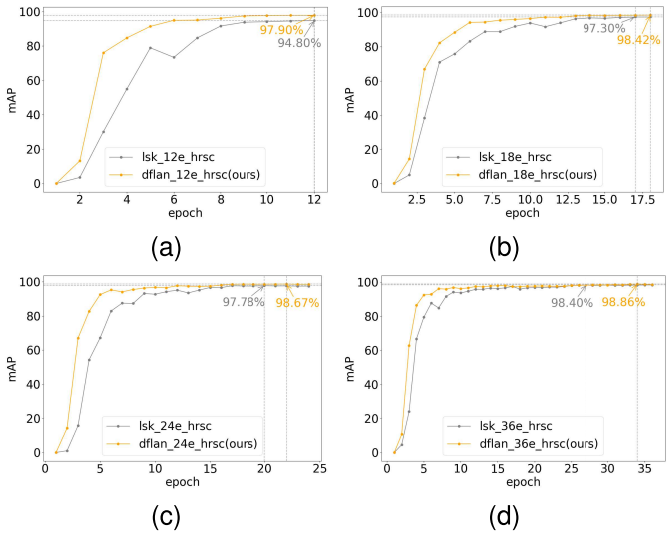


Fig. 12. Relationship between the mAP and epoch under different schedules on the HRSC2016 dataset. (a) 1× training period, 12 epochs. (b) 1.5× training period, 18 epochs. (c) 2× training period, 24 epochs. (d) 3× training period, 36 epochs.

the 1.5× schedule to 18 epochs, the 2× schedule to 24 epochs, and the 3× schedule to 36 epochs. As illustrated in Fig. 12, our method demonstrates significant improvements in convergence speed and final accuracy across these various training schedules. Under the 1× schedule [Fig. 12(a)], our algorithm achieves 97.90% mAP, outperforming the baseline's 94.80% and converging faster. At 1.5× [Fig. 12(b)], it reaches 98.42% mAP, surpassing the baseline's 97.30%. For the 2× schedule [Fig. 12(c)], the mAP improves to 98.67%, compared to 97.78% for the baseline, demonstrating scalability. Finally, under the 3× schedule [Fig. 12(d)], our method achieves 98.86% mAP, exceeding the baseline's 98.40%. Our method assigns more high-quality positive samples to tiny objects, which not only accelerates model convergence but also achieves higher detection performance.

4) Accuracy and Parameters: We analyze the accuracy–parameter tradeoff of different models on the DOTA-v1.0 dataset through Fig. 13. All experiments use a unified 1024 × 1024 input image size. The results show that the proposed DFLAN method achieves the highest detection accuracy (79.42% mAP) with only about 35.4M parameters, significantly outperforming other comparison models (such as LSKNet's 77.89% mAP/31.0M parameters and GWD's 76.34% mAP/47.4M parameters), verifying its excellent balance between high accuracy and lightweight design.

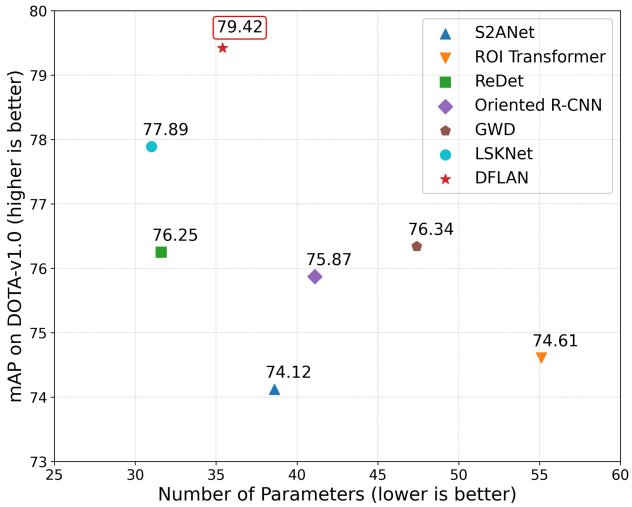


Fig. 13. Comparison of the mAP and parameter count of different models on the DOTA-v1.0 dataset. The DFLAN achieves the best performance.

V. Conclusion

This article proposes a DFLAN for oriented tiny object detection in remote sensing. By incorporating the FSIPN module, our method effectively enhances the features of tiny objects, suppresses background interference, and facilitates efficient communication of multiscale information. To synergistically improve the label assignment accuracy of tiny objects and large objects, we propose the DFLA strategy, which dynamically fuses the IoU and KLD metrics. Additionally, the DHEAD module improves the detection effect of tiny objects through the decoupled head structure and dynamic IoU threshold adjustment mechanism. Extensive experiments on the DOTA-v1.0, HRSC2016, and DIOR-R benchmarks demonstrate the effectiveness of our DFLAN for remote sensing-oriented tiny object detection.

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